

Section 1 Cheat Sheet — Cloud Projects, Accounts & Billing

GCP Study Notes

Study Notes April 2026

Resource Hierarchy

Organization --> Folders (optional, max 10 levels deep) --> Projects --> Resources

- Organization auto-created with Google Workspace or Cloud Identity domain
- Every resource belongs to exactly **one** project
- Policies inherit **downward**; more permissive policy **wins**
- Folders are optional; projects are mandatory

Project Identifiers

Identifier	Unique?	Mutable?	Chosen by
Project Name	No	Yes	User
Project ID	Globally	No (immutable)	User at creation
Project Number	Globally	No	Auto-assigned

Organization Policies

- Control **what** can be done (IAM controls **who**)
- Applied at org/folder/project level, inherited downward

Must-Know Constraints

Constraint	Purpose
<code>compute.vmExternalIpAccess</code>	Block external IPs on VMs
<code>iam.allowedPolicyMemberDomains</code>	Restrict IAM to specific domains
<code>gcp.resourceLocations</code>	Restrict resource regions/zones
<code>compute.trustedImageProjects</code>	Limit allowed boot images
<code>storage.uniformBucketLevelAccess</code>	Force uniform bucket access

IAM Quick Reference

Model: Who (member) + What access (role) + Which resource

Role Types

Type	Scope	Use When
Basic (Primitive)	Broad (viewer/editor/owner)	Almost never in production
Predefined	Service-specific	Default choice
Custom	User-defined permissions	Predefined roles too broad

Basic Roles

Role	Can edit resources?	Can manage IAM?	Can manage billing?
Viewer	No	No	No
Editor	Yes	No	No
Owner	Yes	Yes	Yes

Member Types

`user:` / `serviceAccount:` / `group:` / `domain:` / `allAuthenticatedUsers` / `allUsers`

Best practice: Grant roles to **groups**, not individuals.

Cloud Identity

- IDaaS: user mgmt, groups, SSO, MFA
- **GCDS** syncs from Active Directory/LDAP to Cloud Identity

- Free edition = basic mgmt; Premium = device mgmt + advanced security
- Google Workspace includes Cloud Identity

APIs

- Most APIs **disabled** by default in new projects
- Enabling an API is **free** – usage incurs charges

Quotas

Type	Resets?	Example
Rate quota	Yes (time-based)	API calls per 100s
Allocation quota	No (hard cap)	Max VMs per project

- Applied at **project** level
- Quota exceeded = HTTP **429**
- Request increases via Console: IAM & Admin > Quotas

Operations Suite (formerly Stackdriver)

Product	Purpose
Cloud Monitoring	Metrics, dashboards, alerts, uptime checks
Cloud Logging	Centralized logs, Log Router (to GCS/BQ/Pub/Sub)
Cloud Trace	Distributed tracing / latency
Cloud Profiler	CPU & memory profiling
Error Reporting	Error aggregation

Billing

Core Rules

- Every project must be linked to a billing account to create billable resources
- 1 project → 1 billing account (at a time)
- 1 billing account → many projects
- Billing accounts live at **org level**, outside the folder/project hierarchy

Account Types

Type	Payment
Self-serve (online)	Auto-charged to card/bank
Invoiced (offline)	Monthly invoice, requires approval

Billing Roles

Role	Key Ability
<code>billing.creator</code>	Create new billing accounts
<code>billing.admin</code>	Full billing mgmt (costs, payments, link/unlink)
<code>billing.user</code>	Link projects to a billing account
<code>billing.viewer</code>	View-only costs
<code>billing.costsManager</code>	Manage budgets & exports (no payment mgmt)

To link a project: need `billing.user` on billing account **AND** Project Owner or Project Billing Manager on the project.

Unlinking Billing from a Project

- VMs **stopped**, GCS buckets **read-only**, most services stop
- Resources **not immediately deleted** (grace period)
- Re-link restores billing; stopped resources must be restarted **manually**

Budgets & Alerts

- Default thresholds: 50%, 90%, 100%
- Can be based on **actual** or **forecasted** spend
- Scoped to: billing account / projects / services / labels
- Notifications: email (default), Cloud Monitoring channels, **Pub/Sub**
- Auto-cap spending: Budget Alert → Pub/Sub → Cloud Function → disable billing

Billing Export

Destination	Detail	Use Case
BigQuery (recommended)	Line-item / resource-level	Analysis, dashboards
Cloud Storage	Daily summary CSV/JSON	Archival, legacy

- BigQuery export types: Standard usage cost, Detailed usage cost, Pricing data
- Export is **free** (pay only for BQ queries)
- No backfill – starts from enablement date

Discounts

Type	Commitment	Savings	How
Committed Use (CUD)	1yr or 3yr	Up to 57% / 70%	Manual purchase
Sustained Use (SUD)	None	Up to 30%	Automatic if usage > 25% of month

- Labels (`key:value`) on resources enable cost filtering in billing reports

Essential gcloud Commands

```
# --- Projects ---
gcloud projects list
gcloud projects create PROJECT_ID --name="Name" --folder=FOLDER_ID
gcloud config set project PROJECT_ID
gcloud projects delete PROJECT_ID # 30-day recovery window

# --- Folders / Org ---
gcloud organizations list
gcloud resource-manager folders list --organization=ORG_ID
gcloud resource-manager folders create --display-name="Name" --organization=ORG_ID

# --- Org Policies ---
gcloud org-policies list-available-constraints --organization=ORG_ID
gcloud org-policies reset constraints/CONSTRAINT --organization=ORG_ID

# --- IAM ---
gcloud projects add-iam-policy-binding PROJECT_ID \
  --member="user:a@b.com" --role="roles/viewer"
gcloud projects remove-iam-policy-binding PROJECT_ID \
  --member="user:a@b.com" --role="roles/viewer"
gcloud projects get-iam-policy PROJECT_ID

# --- APIs ---
gcloud services enable compute.googleapis.com
gcloud services list --enabled
gcloud services disable compute.googleapis.com

# --- Billing ---
gcloud billing accounts list
gcloud billing projects link PROJECT_ID --billing-account=BILLING_ID
gcloud billing projects unlink PROJECT_ID
gcloud billing projects describe PROJECT_ID
```

Traps & Gotchas

- **Budgets do NOT cap spending.** They only notify. To auto-cap: Pub/Sub + Cloud Function.
- **Project deletion has a 30-day recovery window** – it is not instant.
- **More permissive policy wins.** You cannot restrict access at a child by removing a parent grant.
- **Enabling an API does not incur charges** – only using the service does.
- **Billing account is NOT inside the folder/project hierarchy** – it sits alongside at the org level.
- **BigQuery billing export has no backfill.** Historical data before enablement is lost.

- **Unlinking billing stops VMs but does NOT delete them immediately.**
- **SUDs are automatic; CUDs require a manual purchase commitment.**
- **Quota errors = HTTP 429.** Quotas are per-project, not per-org.
- `allUsers` = **the entire internet.** Extremely dangerous for IAM bindings.
- **Project ID is immutable.** Choose carefully at creation time.